When and How Might You Die? Visualizing Your Age and Cause of Death Using UN and WHO Mortality Data

Iris Liu

I. Introduction & Motivation

A. Problem Definition and Motivation

When people think about "when and how will I die," they often look at population-level statistics such as life expectancy and overall cause-of-death distributions. However, these numbers are often too abstract and feel distant from individual experience.

Moreover, these numbers do not fully convey the actual risks individuals face. Life expectancy reduces an entire lifetime to a single number, so it does not show how the risk of death changes across different ages. In reality, the risk can vary significantly at different stages of life. Similarly, overall cause-of-death statistics do not show which causes are more relevant at specific ages.

This project uses population-level data to simulate a person's possible age and cause of death. By visualizing these outcomes, it makes mortality statistics more concrete and easier to relate to, and also shows how risks can differ at different stages of life.

B. Target Users and Use Case

This project is designed for a general audience.

It is useful for people who want to understand what population mortality statistics actually mean for them as an individual. It also helps users see how different risks (causes of death) change across ages, which can provide intuition about prevention and health awareness.

II. Related Work

There are many existing visualizations of mortality statistics. For example, https://ourworldindata.org/grapher/life-expectancy shows life expectancy trends across regions and over time. It helps users understand long-term changes, such as improvements in lifespan or the impact of events like wars or pandemics. However, this type of visualization presents data at a macro level and does not show what might happen to an individual today.

The work https://flowingdata.com/2015/09/23/years-you-have-left-to-live-probably/ by Nathan Yau inspired this project. It simulates many possible lives based on mortality statistics and shows the range of ages at which a person might die. Users select their age and gender, and the system generates many outcomes to visualize uncertainty.

This project builds on that idea. First, it adds location (where the user lives), so the simulation better reflects real-world differences across regions. Second, it uses WHO data to assign a cause of death to each simulated life based on age, gender, and region. As a result, the visualization shows not only when a person might die, but also how. This makes it more informative and closer to real-world conditions.

III. Data Description

A. Datasets

The data used in this project come from United Nations global mortality statistics and World Health Organization causes of death statistics

B. Preprocessing

Python was used to extract and clean the data. The UN data were processed to obtain the probability of dying within the next year for individuals aged 1–100, broken down by gender and region, with values provided at 5-year intervals. The WHO data were processed to obtain cause-of-death distributions for each age group, gender, and region.

C. Challenges

The UN and WHO datasets use different definitions of world regions. WHO groups the world into five regions, while UN uses more detailed regional categories.

To make the data consistent, some UN regions were combined and averaged to match the WHO regions. This step was necessary to connect the probability of death with the corresponding causes of death.

IV. Design & Methodology

A. Visualization Design

The visualization is built as an interactive dashboard, as shown in Figure 1.

Users first select their gender, current age, and country at the top.

Based on these inputs, a line chart at the top shows the survival probability across ages.

An animated simulation runs on this line chart. A moving dot represents an individual life. It starts at the current age, moves along the survival probability curve, and falls when death occurs, which is determined by the survival probability at that age. After a dot "dies," a cause of death is assigned

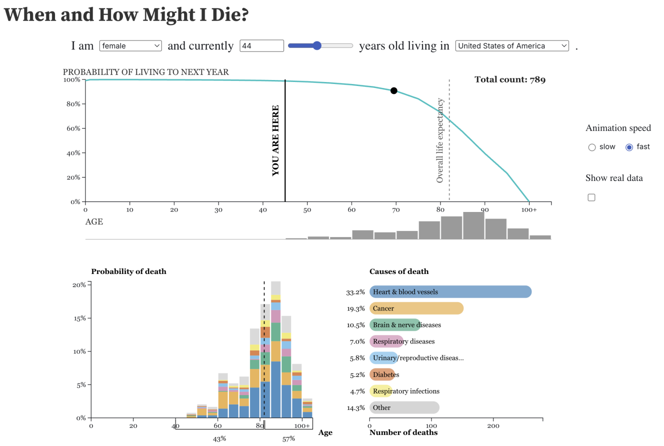

Fig. 1: Overall view of the dashboard.

according to the probability distribution for that specific gender, region, and age of death. A new dot then appears, and the process repeats.

The bar chart at the bottom left shows the distribution of simulated death ages, with stacked colors showing the breakdown by causes of death.

The bar chart at the bottom right shows the overall frequency of causes of death observed in the simulation.

An option to show real data is also provided. When enabled, the visualization overlays the actual statistics on top of the simulation results, allowing users to compare simulated outcomes with the underlying data. This conveys the true likelihood of different outcomes and makes it clear that any single simulated outcome is not deterministic but occurs with a certain underlying probability. As the simulation runs, users will see that the distribution of simulated outcomes gradually converges to the distribution suggested by the real data, as shown in Figure 2.

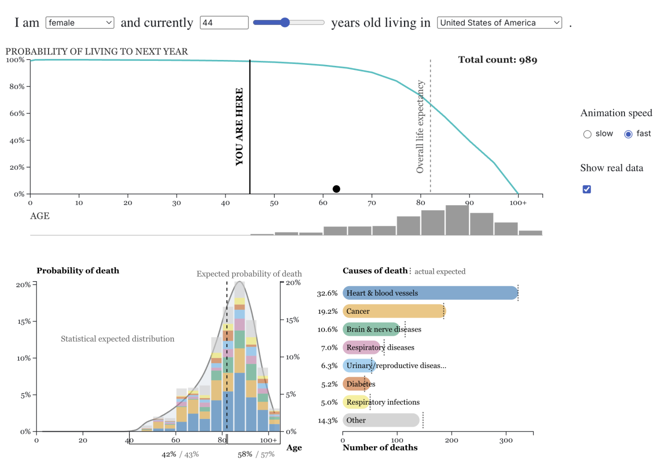

Fig. 2: Simulated distribution eventually converge to real distribution

TABLE I: Visual Encoding Table

Data Variable	**Data Type**	**Mark**	**Channel**
Age	Quantitative	Line	X-Position
Death/Survival Prob.	Quantitative	Line / Bar	Y-Position
Death Count	Quantitative	Bar	X/Y-Position
Cause of Death	Categorical	Bar	Color (hue)

B. *Design Justification*

Table I shows the visual encodings for variables.

Position is used to encode the most important variables (age, death/survival probability and death count), following the Effectiveness Principle.

Color is used to encode the categorical variable of causes of death. A colorblind-safe palette (Okabe-Ito) is used. The same cause of death has the same color across charts.

Animation is used to show the trajectory of each simulated life, and all three charts update in real time as the simulation runs. This makes each "hypothetical life" easy to follow.

C. *Alternative Designs*

Simply visualizing the statistics without simulation was considered. However, simulating the life trajectory and death of hypothetical individuals, and showing this process, helps users see these outcomes as real possibilities rather than just abstract statistics. This makes the simulation results (and eventually the underlying statistics) more concrete and relatable.

V. System Description

A. *Tools*

Python was used for data preprocessing.

Observable was used for interactive visualization and animation.

B. *Architecture*

The system loads two processed CSV files, the UN-based life table data and the WHO cause-of-death probability data. The life table data are filtered according to the user's selected gender and country. The selected country is first mapped to its corresponding WHO region to match the region in UN mortality data and WHO cause-of-death data.

Based on the selected age, gender, and region, the system draws the survival probability curve and identifies the current age point. It also computes the expected probability distribution of death ages from the underlying life table probabilities. This expected distribution is later used when the user enables the "Show real data" option.

The simulation then runs continuously. For each simulated life, a point starts from the user's selected current age and moves along the survival probability curve. As the point moves across age intervals, the system uses the probability of surviving to the next age interval to randomly determine whether the simulated life survives or dies. If death occurs, the point falls from the curve and disappears.

After each simulated death, the system records the death age and samples a cause of death from the WHO cause-of-death distribution that matches the simulated death age group, gender, and region. The recorded result is then added to an internal death record store. The visualization updates after each simulated death.

C. Key Features

- The system includes controls that allow users to select gender, region, and current age. This makes the simulation more personalized and closer to the user's real situation. Users can also explore other possibilities by freely adjusting these settings.
- The system allows users to choose the simulation speed (slow/fast). Slow mode makes it easier to track the trajectory of simulated lives, while fast mode is useful when users want to see the accumulated results of many simulations.
- Linked brushing is enabled. Selecting a cause of death highlights the corresponding data in all related charts (see Figure 3 for an example).

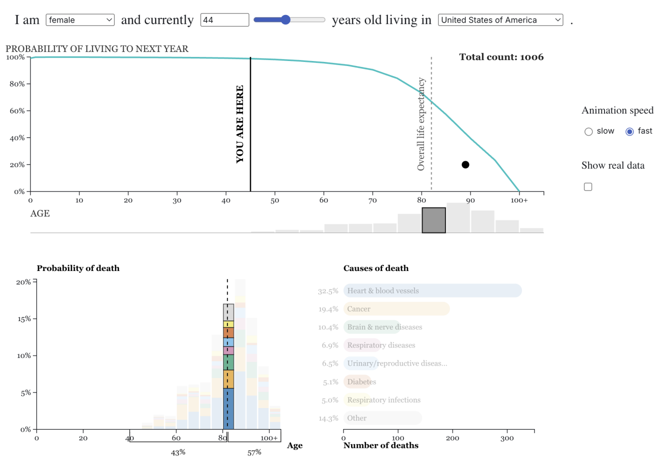

Fig. 3: Selecting an age group highlights the corresponding age group in the related chart

- The system supports hovering (see Figure 4 for an example). Additional details are shown on demand, such as the exact percentage and count of deaths for a given cause and age group.
- The simulation pauses during hover or when a component is highlighted. This allows users to inspect the selected result without the display changing. It resumes automatically once no component is hovered over or highlighted.
- When "Show real data" is turned on, the system overlays the expected statistical distribution on top of the simulated results. As shown in Figure 2, the bottom-left chart displays the expected distribution as a reference curve and shaded area, and the bottom-right chart shows the expected count for each cause as dashed reference lines.
- When "Show real data" is turned on and a specific cause of death is highlighted, the bottom-left chart shows the true probability distribution for that cause (see Figure 5).

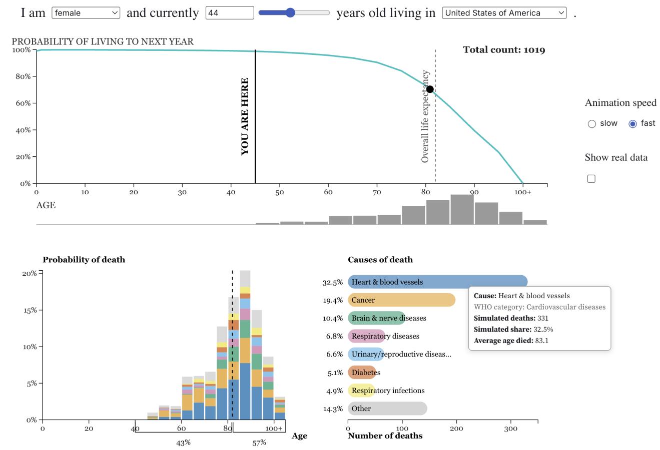

Fig. 4: Hover shows details

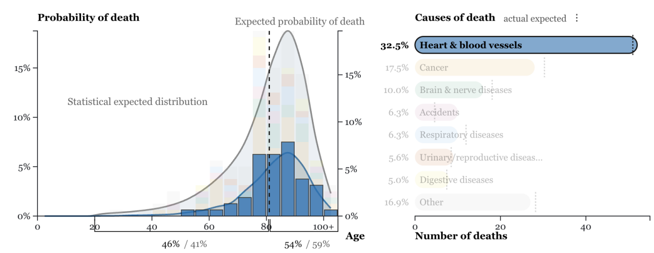

Fig. 5: True distribution of causes of death

- The bottom-left chart includes summary brackets under the x-axis (age) that show the percentage of deaths occurring before and after life expectancy (see Figure 6). When "Show real data" is turned on, the corresponding true percentages from the underlying statistics are displayed next to the simulated results (see Figure 7).

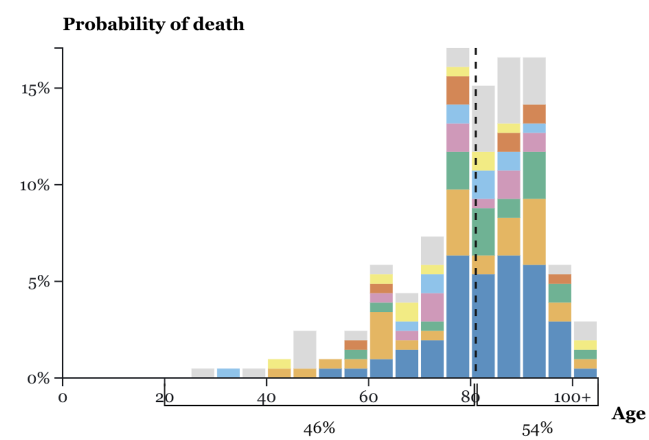

Fig. 6: Percentage of simulated outcomes occurring before and after life expectancy

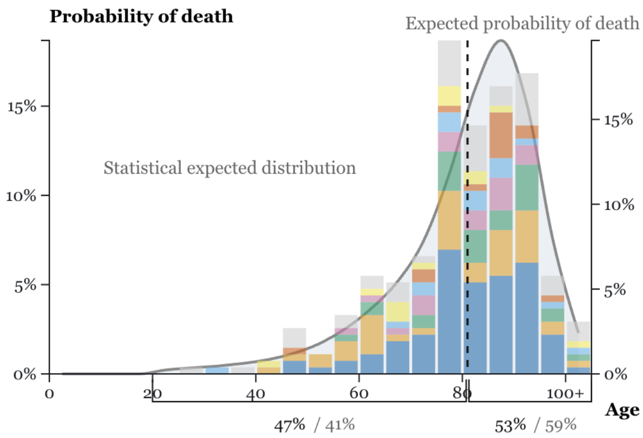

Fig. 7: Percentage of outcomes before and after life expectancy based on the real statistics

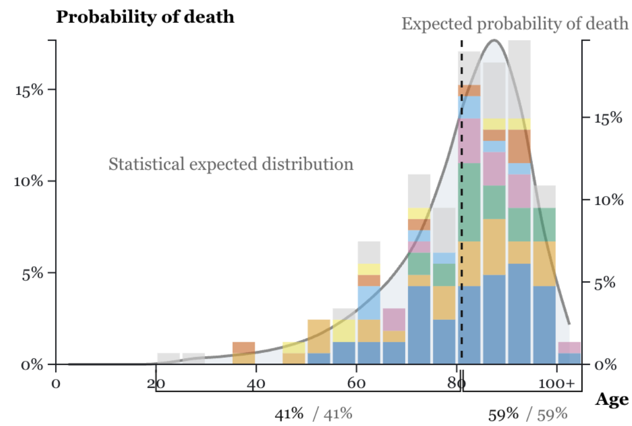

Fig. 9: Large probability of dying before life expectancy

VI. Results & Insights

Observing the simulation results and comparing them with the actual data leads to the following results and insights:

- The (simulated and actual) ages of death are spread over a wide range, rather than concentrating on a single value. An example outcome of the simulation is shown in Figure 8. This means that a person can still die much earlier or much later than the life expectancy.

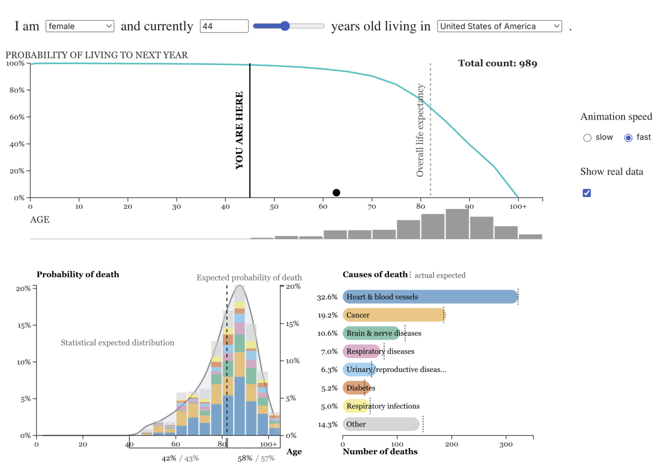

Fig. 8: Sample result

- The peak in both simulated and actual death probability can differ from life expectancy. Figure 8 shows an example where the peak occurs after life expectancy. This suggests that the ages with the highest risk of death do not necessarily coincide with life expectancy.
- There can be a substantial probability of dying before life expectancy. Figure 9 shows an example using my own input, where the result indicates a 41% chance of dying before life expectancy. This is much higher than I expected before starting this project.

- Causes of death vary with age. Some causes appear across almost all ages, while others are concentrated in specific stages of life. For example, under a given setting, Figure 10 shows that infectious and parasitic diseases affect a wide range of ages, while Figure 11 shows that brain and nerve diseases are concentrated in a narrower age range, mainly in later life.

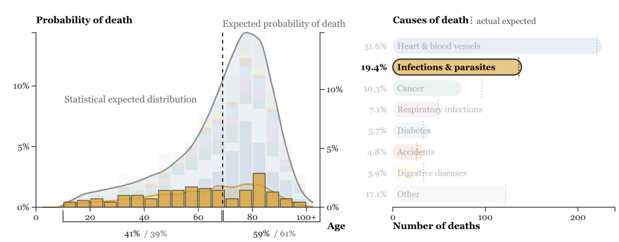

Fig. 10: Some causes affect a wide range of ages.

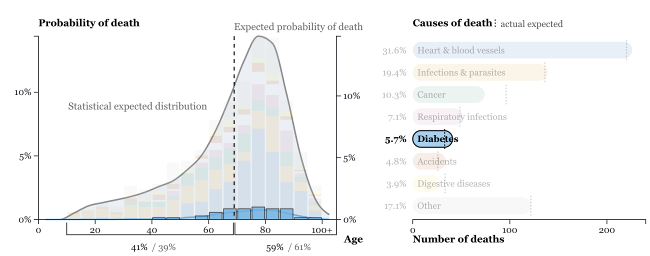

Fig. 11: Some causes are concentrated in later stages of life.

- Although the outcome of each simulated life (when it dies and how it dies) is inherently uncertain and cannot be predicted, individual outcomes add up to form a distribution that matches the overall statistical pattern, which is a "certain" result, as can be seen in Figure 2. This reflects how uncertainty and certainty intertwine in real life.

VII. Limitations & Future Work

- There are many causes of death in the data, but it is not practical to display all of them clearly in the visualization. As a result, less frequent causes are combined into an "other" category. This makes the chart easier to read, but it can also lead to a very large 'other" bar, which hides useful details. In the future, this could be improved by allowing users to expand the 'other" category or view a more detailed breakdown on demand.
- Results are not truly country-specific. Although users can select a country, the underlying data only exist at the level of the five WHO regions. Each country is mapped to one of these regions, so the results do not reflect true country-specific differences. In the future, this could be improved by using country-level mortality data if available.
- The latest WHO mortality data record COVID-19 as a major cause of death across all regions. Since this no longer reflects current conditions, 2019 statistics are used instead. In the future, when post-COVID data become available, they may better capture both the longer-term effects of COVID-19 on health outcomes and more recent mortality patterns.

VIII. Conclusion

This project explores how to make mortality statistics more meaningful at the individual level. Instead of showing only abstract numbers such as life expectancy or overall cause-of-death distributions, the visualization uses simulation to show what these statistics look like for a single "hypothetical life."

The results show that life expectancy is only a reference point, while actual outcomes can vary over a wide range. The visualization also reveals how causes of death change across different ages. They help users better understand both the uncertainty and the structure in mortality data. Finally, by comparing the simulation results with the underlying statistical data, the visualization further shows how individual uncertainty accumulates into stable population-level patterns. While each simulated life follows a different trajectory, the overall distribution matches the expected probabilities. This comparison helps fill the gap between individual experience and population statistics.